

XRPL Validator — Daily Engineering Report • 2026-08-30

Compiled retrospectively on 2026-09-23 from the git history. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the XRP Ledger itself (amendments) and XRPL research.

Summary

- 5 commits (xrpl-validator 5)

Engineering changes

- **fix(xrpl-ledger): XRP delivery re-arms the destination's free key reset** (99bd44a , xrpl-validator) Finding 35 (#106644275, dst rMWEptMt): receiving an XRP payment clears the destination's IsfPasswordSpent — 'Re-arm the password change fee if we can and need to', Payment.cpp:674-676, pinned by SetRegularKey_test.cpp:98-100 (fee-0 regkey sets the flag, a later inbound payment clears it). Ours never cleared, and PasswordSpent accounts receive XRP constantly: the fresh window cascaded from ledger two. 2/61 -> 46/60; ...
- **fix(xrpl-ledger): LP deposit sum rounds half-even at 16 digits** (ba4ae17 , xrpl-validator) Finding 36 (#106644320 749D3E45, tfTwoAsset): the minted LP amount was already byte-exact — the divergence was THE ADD. The exact sum 279228264.81815046 must 16-digit-round nearest (truth ...1505); both the old exact-string path and a downward round land ...1504. Receipts (new DX_AMMDEP: minted/lp_pre/lp_post/sized arm) pinned it after two model-driven theories missed: the revival-initializer port (kept — correct per ...
- **feat(scripts): durable VHS listing watch (systemd user timer, 30min)** (1af1d6d , xrpl-validator) We are absent from the VHS validator list (manifest rejected: master == ephemeral). Session-bound watches die silently, so a cron-style timer polls data.xrpl.org and telegrams once per state change: first listing, server_version rendering, or de-listing. Timer offset :07/:37 keeps clear of the 03:00 batch window.
- **fix(ledger): regross the want pot from the net remainder each iteration (finding 37)** (82df826 , xrpl-validator) rippled's remainingOut lives NET (savedOuts in outReq units) and every flow iteration regrosses it through the last DirectStep's rev pass with mulRatio roundUp. Our gross-subtraction pot (want_gross minus grossed fills) cancels tails the fresh product keeps: #106644326 431CAAFC iter-5 net 1.442557458781219 x 1.001 = ...240000|219 regrosses to ...240001 while the pot said ...240000, and the UNI/XRPS book rev-sizing ...
- **feat(node): shadow hydration v5 memory diet (budget gate, no-cache scan, RSS receipts)** (2bb284f , xrpl-validator) v4's single greedy pass spiked live_viewer to 58.6GB on the 62GB box and the machine thrashed; the mirror itself (~13-15GB) fits — the SPIKE is the killer. Three levers: - Budget gate: hydrate refuses to start unless /proc/meminfo MemAvailable >= XRPL_SHADOW_HYDRATE_MIN_GB (default 20). A skipped hydrate is a receipt (hydrate_skipped_lowmem stat + log line) and re-arms the 600s cooldown; a thrashing validator is an ...

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Cooling-Off Custody Withdrawal Holds** (custody) Add an AccountPolicy ledger object (ltACCOUNT_POLICY, owner-reserved, keylet::accountPolicy(account)) set via a new AccountPolicySet transaction carry
- **Deterministic Jittered Iceberg Replenishment** (dex) Extend g31's tfIceberg OfferCreate flag by replacing the fixed refill size with sfMinReplenish and sfMaxReplenish SFields alongside sfDisplayQuantity.
- **Bounded WASM Hooks for Mainnet** (hooks) Port Xahau's proven Hooks execution model to XRPL mainnet as the first concrete implementation of the in-flight XLS-101/XLS-102 line: add a Hook ledge

Appendix — commits

Time	Repo	Commit	Subject
09:35	xrpl-validator	99bd44a	fix(xrpl-ledger): XRP delivery re-arms the destination's free key reset
10:54	xrpl-validator	ba4ae17	fix(xrpl-ledger): LP deposit sum rounds half-even at 16 digits
12:52	xrpl-validator	1af1d6d	feat(scripts): durable VHS listing watch (systemd user timer, 30min)
12:52	xrpl-validator	82df826	fix(ledger): regross the want pot from the net remainder each iteration (finding 37)
13:01	xrpl-validator	2bb284f	feat(node): shadow hydration v5 memory diet (budget gate, no-cache scan, RSS receipts)